GE VERNOVA

GridOS SCADA

0.7.0

Metrics User Guide

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Change History

Date	Change Description
2025-Mar-31	GridOS SCADA 0.1.0: • This is a new document for GridOS SCADA 0.1.0.
2025-Jul-04	GridOS SCADA 0.4.0: • Removed anybase connector section.
2026-March-03	GridOS SCADA 0.7.0: • Updated SCADA Service Metrics section with new counters *Analog Process Alarm Counter* and *Status Process Alarm Counter*.

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1. GridOS SCADA Metrics Introduction

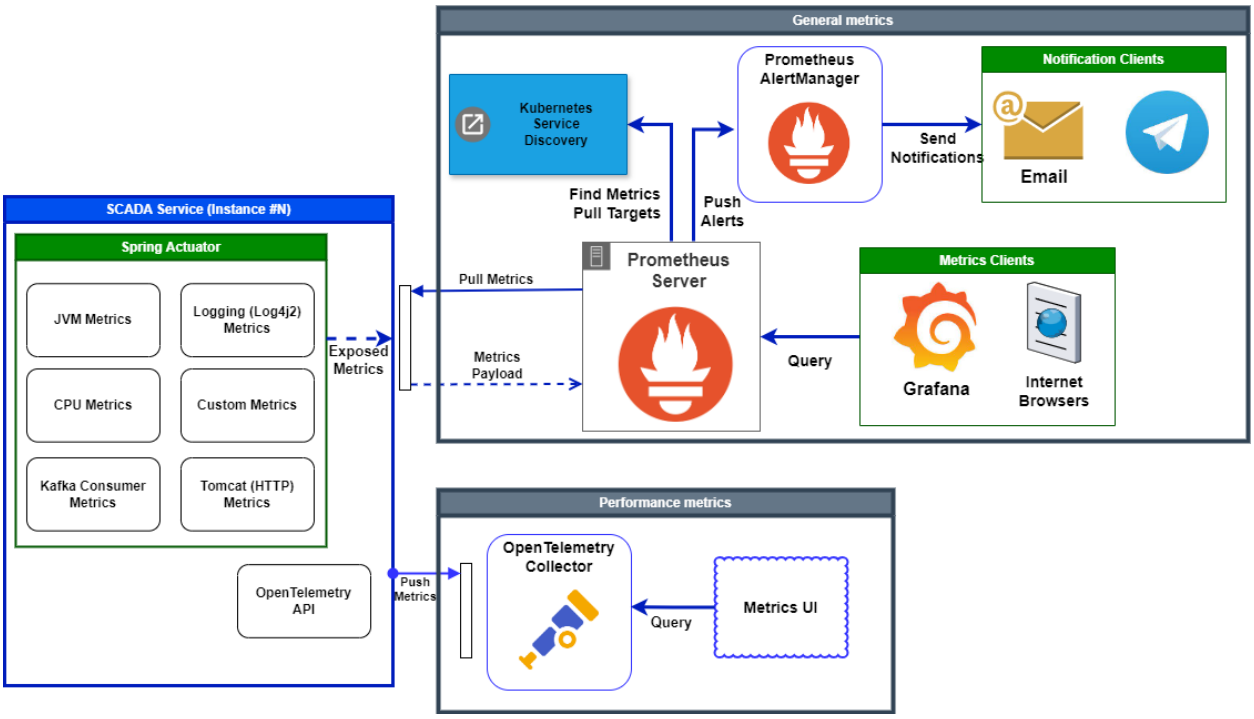


Figure 1. Metrics Data Flow Diagram

The metrics tools (Spring Actuator and OpenTelemetry) will send the metrics data (push or pull) to their respective collectors (Prometheus Server and OpenTelemetry Collector). The collected data is used by Grafana to process, group, and display the data.

1.1. Metrics

The metrics will be collected using the Spring Boot Actuator module, which comes with various APIs and endpoints that support numerous monitoring systems.

Currently, the GridOS SCADA Service and Anybase connectors use Prometheus to get metrics.

There are built-in metrics that are exposed by default by the Spring Boot Actuator, and there are also additional custom metrics added to the GridOS SCADA Service and Anybase connectors.

Metrics need to be registered with the Micrometer library, and then Spring Boot Actuator can collect metrics when a client (for example, Grafana) polls the data at the Prometheus endpoint.

1.2. Viewing Metrics via Grafana

Prerequisite:

- Configure Grafana as described in Accessing Monitoring Web Applications.

To view the SCADA metrics:

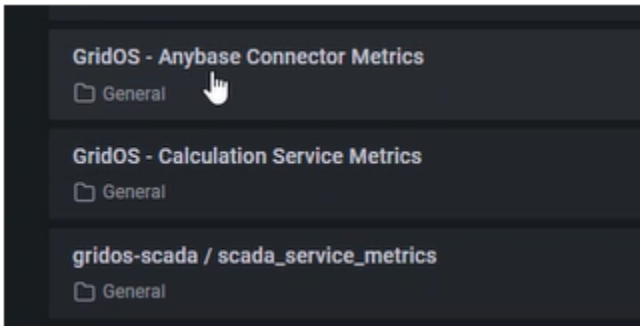
1. Access Grafana.

   If you have configured port forwarding, enter `localhost:3000` in the URL bar of a browser.

   > **Tip**
   >
   > ***GE Vernova internal:*** You can alternatively use the link provided by ingress_links.html through the PTE build (for example, `https://scada-·····.amazonaws.com/monitoring/grafana`)

   The Grafana home page appears.

2. In the Dashboards section, select one of the provided GridOS SCADA dashboards:

   The dashboard appears.

   > **Tip**
   >
   > For information about using the dashboards controls, refer to Grafana documentation (https://grafana.com/docs/grafana/latest/dashboards/use-dashboards/)

2. GridOS SCADA Service Metrics

2.1. Custom Metrics

- **Init App Gauge:** Application initialization time.

  This metric is used to measure the time from application start to when the application is in a ready state, which includes the initialization of all tables, fields, and so on.

- **Analog Update Recv Counter:** Total number of received Analog updates.

  This metric is used to measure the number of incoming analog update requests since the application started.

- **Status Update Recv Counter:** Total number of received Status updates.

  This metric is used to measure the number of incoming status update requests since the application started.

- **Accum Update Recv Counter:** Total number of received Accumulator updates.

  This metric is used to measure the number of incoming accumulator update requests since the application started.

- **Limit Update Recv Counter:** Total number of received Limit updates.

  This metric is used to measure the number of incoming limit update requests since the application started.

- **Dead Letter Queue Counter:** Total number of messages sent to the Dead Letter Queue.

  This metric is used to measure the number of incoming update requests that failed and were sent to the dead letter queue.

- **Accum Processing Msg Timer:** Time to process incoming Accumulator messages.

  This metric is used to measure how long it takes to process an accumulator update request. It contains a count, which is the number of incoming accumulator update requests since the application started, and a summation, which is the total processing time of all these messages.

- **Analog Processing Msg Timer:** Time to process incoming Analog messages.

  This metric is used to measure how long it takes to process an analog update request. It contains a count, which is the number of incoming analog update requests since the application started, and a summation, which is the total processing time of all these messages.

- **Status Processing Msg Timer:** Time to process incoming Status messages.

  This metric is used to measure how long it takes to process a status update request. It contains a count, which is the number of incoming status update requests since the application started, and a summation, which is the total processing time of all these messages.

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- **Limit Processing Msg Timer:** Time to process incoming Limit messages.

  This metric is used to measure how long it takes to process a limit update request. It contains a count, which is the number of incoming limit update requests since the application started, and a summation, which is the total processing time of all these messages.

- **Write Latency Timer:** Micrometer timer to track how long it takes for the Anybase writing process for the requests.

  The timer contains a sum component to keep track of the total write processing time of all the update requests and a counter to keep track of the number of update requests.

- **Analog Process Alarm Counter:** Total number of alarms generated during the processing of incoming Analog updates.

  This metric is used to measure the number of alarms generated during the processing of incoming Analog updates.

- **Status Process Alarm Counter:** Total number of alarms generated during the processing of incoming Status updates.

  This metric is used to measure the number of alarms generated during the processing of incoming Status updates.

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